

# Zcash, end to end

from a wallet seed to a verified shielded payment

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**One order**

## Why nodes must agree on an order

Two transactions spend the same coin. Each is valid alone; together, one must lose.



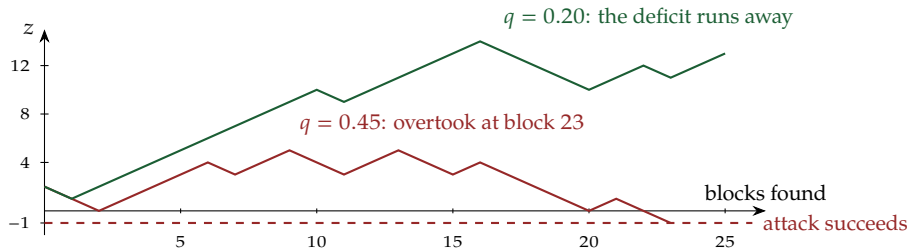
Consensus supplies a **canonical order**: the chain with the most work. Every later rule, the note tree, the nullifier set, the pool balances, is a function of that order.

# The race

An attacker mines a secret branch;  $z$  is the honest chain's lead in blocks.

$$z_{i+1} = \begin{cases} z_i + 1 & \text{honest block, probability } p \\ z_i - 1 & \text{attacker block, probability } q = 1 - p \end{cases}$$

The attack succeeds when  $z$  reaches  $-1$ .



## Deriving the catch-up probability

Let  $h_N(z)$  be the probability that the walk from  $z$  reaches  $-1$  before a ceiling  $N$ .  
Condition on the next block:

$$h_N(z) = p h_N(z+1) + q h_N(z-1), \quad h_N(-1) = 1, \quad h_N(N) = 0.$$

The trial solution  $h_N(z) = t^z$  works exactly at the roots of

$$p t^2 - t + q = p (t-1) (t - q/p),$$

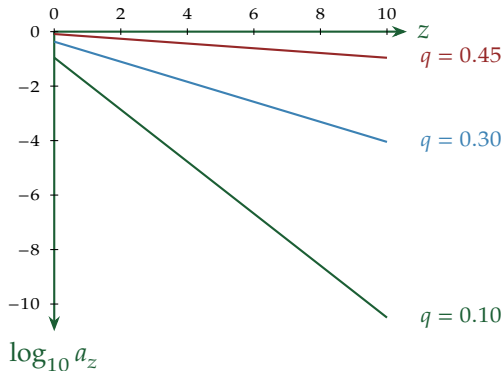
so for  $q \neq p$ ,  $h_N(z) = A + B (q/p)^z$ , and the two boundaries give

$$h_N(z) = \frac{(q/p)^z - (q/p)^N}{(q/p)^{-1} - (q/p)^N} \xrightarrow{N \rightarrow \infty} a_z = (q/p)^{z+1} \quad (q < p).$$

For  $q \geq p$  the limit is 1: an attacker with half the hashrate or more always catches up.

# The catch-up probability

$$a_z = \Pr[\text{the walk ever reaches } -1 \text{ from } z] = \begin{cases} (q/p)^{z+1} & q < p \\ 1 & q \geq p \end{cases}$$



| $q$  | $a_0$ | $a_5$                |
|------|-------|----------------------|
| 0.10 | 0.111 | $1.9 \times 10^{-6}$ |
| 0.30 | 0.429 | $6.2 \times 10^{-3}$ |
| 0.45 | 0.818 | 0.30                 |

Each block of lead multiplies  $a_z$  by  $q/p$ : the order of old blocks becomes final exponentially fast.

## What the order gives the shielded protocol

| object                      | a function of the ordered chain                            |
|-----------------------------|------------------------------------------------------------|
| note commitment tree anchor | every cmx, appended in order the tree's root after a block |
| nullifier set               | every nf published so far                                  |
| pool balance                | minus the sum of every $v_{\text{balance}}$                |

From here on: one canonical chain. The rest of the talk follows one payment through it.

# Primitives

# Groups and generators

| curve                                                | equation                                              | group order         |
|------------------------------------------------------|-------------------------------------------------------|---------------------|
| Pallas $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ | $y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\text{Pallas}}}$ | $p_{\text{Vesta}}$  |
| Vesta                                                | $y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\text{Vesta}}}$  | $p_{\text{Pallas}}$ |

Two primes  $p_{\text{Pallas}}, p_{\text{Vesta}} \approx 2^{254}$ : each curve's scalar field is the other's base field.  
Discrete logarithms cost about  $2^{126}$  group operations.

$$[a]P = P + \dots + P, \quad \text{Extract}_{\mathbb{P}}((x, y)) := x, \quad \text{GroupHash}(D, M) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$$

Every generator is GroupHash of its own label:  $G^{\text{SpendAuth}}$  (spend authorisation),  $K^{\text{Orchard}}$  (nullifiers),  $V^{\text{Orchard}}, R^{\text{Orchard}}$  (values),  $\mathbf{g}_d$  (addresses),  $Q, S(j)$  (Sinsemilla). No relation among them is known, and nothing is set up in trust.

# Pseudorandom functions

For a secret uniform key, outputs look uniform and independent: one secret expands into many keys.

| function                                   | built from                                                                                  | used for          |
|--------------------------------------------|---------------------------------------------------------------------------------------------|-------------------|
| $\text{PRF}_k(t, \dots)$                   | BLAKE2b, into $\mathbb{F}_{p_{\text{Pallas}}}$ or $\mathbb{F}_{p_{\text{Vesta}}}$ if needed | every derived key |
| $\text{PRF}_{\text{nk}}^{\text{nf}}(\rho)$ | Poseidon, over $\mathbb{F}_{p_{\text{Pallas}}}$                                             | nullifiers        |
| $\text{KDF}(s, \text{epk})$                | BLAKE2b                                                                                     | encryption keys   |

The tag  $t$  names the output:  $\text{PRF}_{\text{sk}}(\text{ask})$  says nothing about  $\text{PRF}_{\text{sk}}(\text{nk})$ . The nullifier PRF is  $\text{PRF}_{\text{nk}}^{\text{nf}}(\rho) := \text{PoseidonHash}(\text{nk}, \rho)$ , an algebraic hash the proof can evaluate cheaply.

# The Sinsemilla hash

Generators  $S(0), \dots, S(1023)$  and a base  $Q(D)$  for each domain  $D$ . Split  $M$  into 10-bit chunks  $m_1, \dots, m_n$ :

$$\text{Acc}_0 := Q(D), \quad \text{Acc}_i := (\text{Acc}_{i-1} + S(m_i)) + \text{Acc}_{i-1}$$

$$\text{SinsemillaHashToPoint}(D, M) := \text{Acc}_n = [2^n] Q(D) + \sum_{i=1}^n [2^{n-i}] S(m_i)$$

$$\text{SinsemillaHash}(D, M) := \text{Extract}_{\mathbb{P}}(\text{Acc}_n)$$

- A Pedersen-style hash: two messages of one length with one output give a discrete-log relation among  $Q(D), S(0), \dots, S(1023)$ .
- Each 10-bit step is one table lookup and two additions: cheap to prove.

# Sinsemilla commitments

$$\text{SinsemillaCommit}_r(D, M) := \text{SinsemillaHashToPoint}(D, M) + [r] H_D$$

$$\text{ShortCommit}_r(D, M) := \text{Extract}_{\mathbb{P}}(\text{SinsemillaCommit}_r(D, M))$$

$r \in \mathbb{F}_{p_{\text{Vesta}}}$  uniform;  $H_D$  another hashed generator.

- **Hiding:** for uniform  $r$  the commitment is a uniform point, whatever  $M$ .
- **Binding:** two openings of one commitment give a discrete-log relation.

| instance                                             | form             | input          |
|------------------------------------------------------|------------------|----------------|
| $\text{NoteCommit}_{\text{rcm}}$                     | commitment       | a note         |
| $\text{Commit}_{\text{r}_{\text{ivk}}}^{\text{ivk}}$ | short commitment | (ak, nk)       |
| MerkleCRH                                            | hash             | two tree nodes |

# Value commitments

$$\text{cv}(v, \text{rcv}) := [v] V^{\text{Orchard}} + [\text{rcv}] R^{\text{Orchard}}$$

$v$  a signed amount;  $\text{rcv} \in \mathbb{F}_{p_{\text{Vesta}}}$  uniform.

- **Hiding:** uniform  $\text{rcv}$  gives a uniform point.
- **Binding:** one point opened to two amounts gives  $\log_{R^{\text{Orchard}}} V^{\text{Orchard}}$ .
- **Homomorphic:**

$$\text{cv}(v_1, r_1) + \text{cv}(v_2, r_2) = \text{cv}(v_1 + v_2, r_1 + r_2)$$

The commitments of a bundle add up to a commitment to its net value.

## Key agreement and encryption

The recipient publishes  $(g_d, pk_d)$  with  $pk_d = [ivk] g_d$ . The sender picks a fresh  $esk$ :

$$epk := [esk] g_d, \quad s := [esk] pk_d = [esk \cdot ivk] g_d = [ivk] epk$$

$$K := \text{KDF}(s, epk), \quad C := \text{Enc}_K(\text{plaintext})$$

Enc is authenticated encryption (ChaCha20-Poly1305): decryption under a wrong key fails.

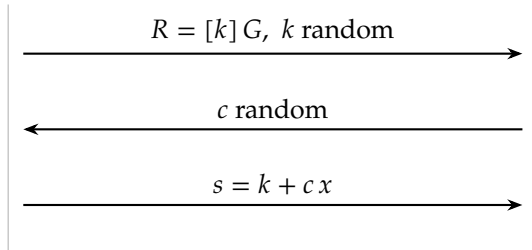
Diffie–Hellman on a base chosen by the recipient:  $s$  needs  $ivk$  or  $esk$ . The pair  $(epk, C)$  names no one.

# A sigma protocol

The prover knows  $x$  with  $X = [x] G$ .

**Prover**

**Verifier**



Verifier checks  $[s] G \stackrel{?}{=} R + [c] X$

$x, k, c, s \in \mathbb{F}_{p_{\text{Vesta}}}$ ;  $G, X, R \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ .

## Why it proves, and why it hides

**Special soundness.** Two accepting transcripts with the same  $R$ :

$$(R, c, s), (R, c', s') \implies x = \frac{s - s'}{c - c'}$$

whoever answers two challenges knows  $x$ .

**Zero knowledge.** A simulator with no  $x$  picks  $c$  and  $s$  first:

$$R := [s] G - [c] X$$

the transcript has exactly the real distribution. The verifier learns nothing it could not have made itself.

## Fiat–Shamir: the verifier becomes a hash

$$c := H^{\odot}(R, X, M)$$

- The prover cannot pick  $R$  after  $c$ :  $c$  depends on  $R$ .
- No interaction: anyone can check  $(R, s)$  later.
- Binding a message  $M$  into  $c$  makes  $(R, s)$  a signature on  $M$ .
- **sigma protocol + Fiat–Shamir = Schnorr signature.**

$H^{\odot}$  is modelled as a random oracle; Halo 2 draws its challenges the same way.

# Schnorr signatures

Key  $x \in \mathbb{F}_{p_{\text{Vesta}}}$ , public  $X := [x] B$ ; to sign  $M$ , pick a fresh  $k$ :

$$R := [k] B, \quad c := H^{\otimes}(R, X, M), \quad s := k + c x$$

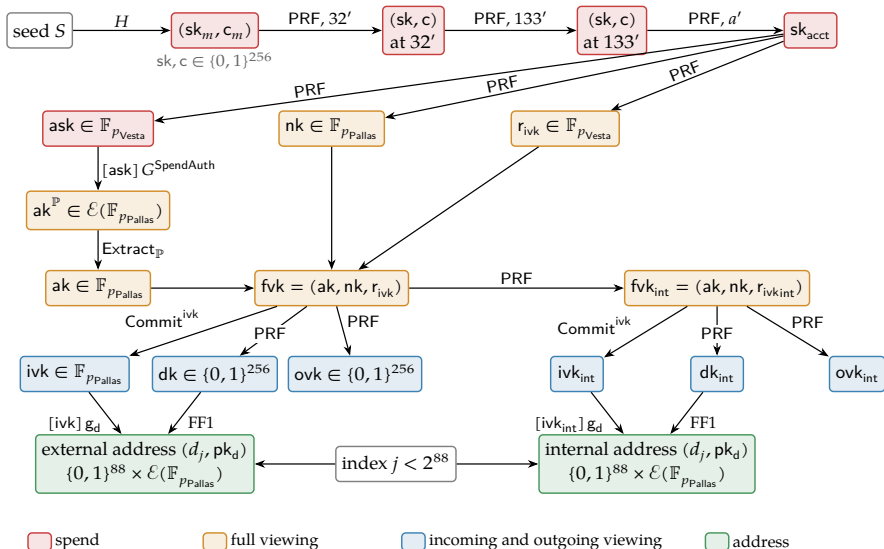
$$\text{verify}(R, s): \quad [s] B = R + [c] X$$

- Re-randomisable:  $(x + \alpha, X + [\alpha] B)$  is a fresh-looking key pair, and signing under it still needs  $x$ .
- RedPallas: spend authorisation on  $B = G^{\text{SpendAuth}}$  with re-randomised keys; the binding signature on  $B = R^{\text{Orchard}}$ .

Unforgeable under discrete logarithms, with  $H^{\otimes}$  a random oracle.

# Keys

# The key structure



## From the seed to an account key

$$(\text{sk}_m, c_m) := H(S), \quad (\text{sk}_i, c_i) := \text{PRF}_{c_{\text{par}}}(\text{sk}_{\text{par}}, i)$$

$S$  the wallet seed;  $H$  a hash with two halves; each chain code  $c$  keys the next step.

Path  $m/32'/133'/a'$ : purpose, coin type, account.

- Hardened only: a child reveals nothing about its parent or its siblings.
- Account  $a$  keeps  $\text{sk} := \text{sk}_{m/32'/133'/a'}$  and drops the chain code.

## Spending secrets

$$\text{ask} := \text{PRF}_{\text{sk}}(\text{ask}) \in \mathbb{F}_{p_{\text{Vesta}}}$$

$$\text{nk} := \text{PRF}_{\text{sk}}(\text{nk}) \in \mathbb{F}_{p_{\text{Pallas}}}$$

$$r_{\text{ivk}} := \text{PRF}_{\text{sk}}(r_{\text{ivk}}) \in \mathbb{F}_{p_{\text{Vesta}}}$$

$$\text{ak}^{\mathbb{P}} := [\text{ask}] G^{\text{SpendAuth}}, \quad \text{ak} := \text{Extract}_{\mathbb{P}}(\text{ak}^{\mathbb{P}})$$

The key  $\text{ask}$  signs spends;  $\text{nk}$  keys nullifiers;  $r_{\text{ivk}}$  blinds the viewing key.

## Viewing keys

$$\text{ivk} := \text{Commit}_{r_{\text{ivk}}}^{\text{ivk}}(\text{ak}, \text{nk}), \quad (\text{dk}, \text{ovk}) := \text{PRF}_{r_{\text{ivk}}}(\text{ak}, \text{nk})$$

| key                                      | lets its holder                    |
|------------------------------------------|------------------------------------|
| ask                                      | spend                              |
| $(\text{ak}, \text{nk}, r_{\text{ivk}})$ | see spends (nullifiers)            |
| $(\text{dk}, \text{ivk})$                | see incoming notes, make addresses |
| ovk                                      | recover what it sent               |

# The address

For an index  $j$ :

$$d := \text{FF1}_{\text{dk}}(j), \quad g_d := \text{GroupHash}(d), \quad \text{pk}_d := [\text{ivk}] g_d$$

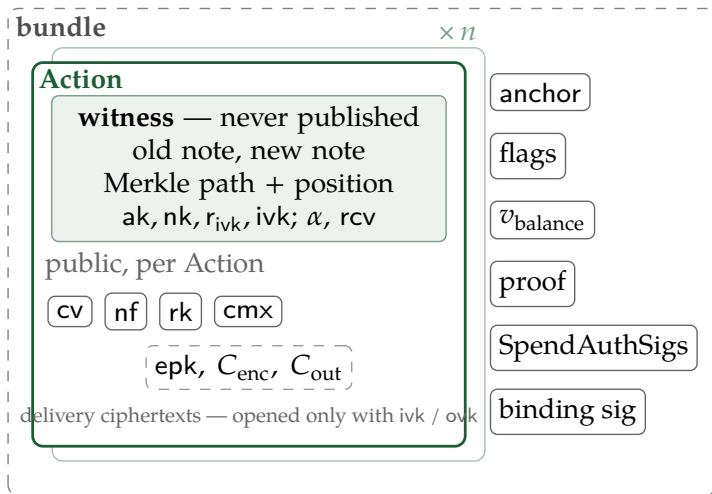
The address is  $(d, \text{pk}_d)$ .

FF1 is a keyed pseudorandom permutation of 88-bit strings.

- $2^{88}$  addresses per key, all with  $\text{pk}_d = [\text{ivk}] g_d$ .
- Two addresses of one key are unlinkable without dk or ivk.
- The address is all a sender needs.

**Someone pays you**

# The bundle



A transaction carries a bundle of  $n$  Actions, with one anchor, one set of flags, one value balance  $v_{balance}$ , one proof and one binding signature.

# The Action

Each Action spends one note and creates one:

| side   | published             | what it is                                                     |
|--------|-----------------------|----------------------------------------------------------------|
| spend  | nf                    | the spent note's nullifier                                     |
|        | rk                    | the key its signature is checked against                       |
|        | $cv^{\text{net}}$     | a commitment to old minus new value                            |
| output | cmx                   | the new note's commitment                                      |
|        | epk, $C_{\text{enc}}$ | the new note, encrypted to its recipient                       |
|        | $C_{\text{out}}$      | $(pk_d, esk)$ under the sender's ovk: reopens $C_{\text{enc}}$ |

An unused side is a dummy note of value 0. No field names a sender, a recipient or an amount.

# Trial decryption

For every Action on the chain, the wallet tries its ivk:

1.  $s := [\text{ivk}] \text{epk}$ ,  $K := \text{KDF}(s, \text{epk})$ ,  $(d, v, \text{rseed}, \text{memo}) := \text{Dec}_K(C_{\text{enc}})$ ;
2.  $\rho := \text{nf}$  of the same Action;  $\text{esk}$ ,  $\psi$  from  $\text{rseed}$  and  $\rho$ ;
3.  $g_d := \text{GroupHash}(d)$ ,  $\text{pk}_d := [\text{ivk}] g_d$ ,  $\text{rcm}$  from  $\text{rseed}$  and  $(g_d, \text{pk}_d, v, \rho, \psi)$ ;
4. accept only if  $[\text{esk}] g_d = \text{epk}$  and  
 $\text{Extract}_{\mathbb{P}}(\text{NoteCommit}_{\text{rcm}}(g_d, \text{pk}_d, v, \rho, \psi)) = \text{cmx}$ .

An Action meant for someone else fails the decryption or step 4. An accepted note is the unique opening of  $\text{cmx}$ : the sender cannot hand over a note the wallet could not later spend.

Light wallets run this on a compact form of each Action:  $\text{nf}$ ,  $\text{cmx}$ ,  $\text{epk}$  and the start of  $C_{\text{enc}}$ , without the memo.

## The note

Decryption yields

$$n = ((d, \text{pk}_d), v, \rho, \psi, \text{rcm})$$

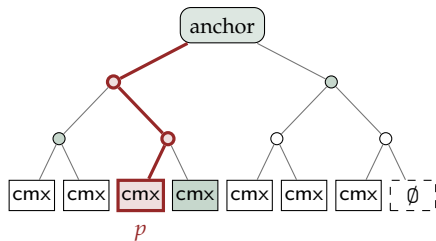
with  $v$  the amount and  $\rho$  the nullifier of the note spent in the same Action. One seed fixes the rest:

$$\begin{aligned} \text{esk} &:= \text{PRF}_{\text{rseed}}(\text{esk}, \rho), & \psi &:= \text{PRF}_{\text{rseed}}(\text{psi}, \rho), \\ \text{rcm} &:= \text{PRF}_{\text{rseed}}(\text{rcm}, g_d, \text{pk}_d, v, \rho, \psi) \end{aligned}$$

$$\text{cm} := \text{NoteCommit}_{\text{rcm}}(g_d, \text{pk}_d, v, \rho, \psi), \quad \text{cmx} := \text{Extract}_{\mathbb{P}}(\text{cm})$$

The chain holds  $\text{cmx}$  in the note commitment tree; the note itself never appears.

## Keeping a witness



The note's cmx sits at position  $\text{pos}$  of a depth-32 tree. With  $h_t := \text{MerkleCRH}(31 - t, \cdot, \cdot)$ :

$$a_0 := \text{cmx}, \quad a_{t+1} := \begin{cases} h_t(a_t, s_t) & b_t = 0 \\ h_t(s_t, a_t) & b_t = 1 \end{cases}$$

Valid path:  $a_{32} = rt$ , the anchor.

Every sibling  $s_t$  is a function of the cmx values appended so far: the wallet updates the path from the blocks it scans and never reveals  $\text{pos}$ .

## Watching for spends

The wallet computes its note's nullifier once:

$$\text{nf} := \text{Extract}_{\mathbb{P}}([\text{PRF}_{\text{nk}}^{\text{nf}}(\rho) + \psi) \bmod p_{\text{Pallas}}] K^{\text{Orchard}} + \text{cm})$$

and marks the note spent when nf appears in a scanned block.

- Deterministic: one note, one nullifier.
- Unlinkable: without nk it looks random.
- The wallet now holds everything a spend needs: the note, pos, the path, an anchor.

**Spending**

## The spend side of an Action

$$\begin{array}{ll} \text{nf}^{\text{old}} := \text{DeriveNullifier}_{\text{nk}}(\rho, \psi, \text{cm}) & \text{published} \\ \text{rsk} := \text{ask} + \alpha, \quad \text{rk} := \text{ak}^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}} & \text{fresh } \alpha \text{ per Action} \\ \text{cv}^{\text{net}} := \text{cv}(v^{\text{old}} - v^{\text{new}}, \text{rcv}) & \text{fresh rcv} \end{array}$$

- rk is a uniform point: two spends by one key are unlinkable.
- $\text{cv}^{\text{net}}$  is a uniform point: the amount is hidden, and commitments add up.

## The output side

Each output is a new note, made as the received one was, with a fresh rseed and

$$\rho^{\text{new}} := \text{nf}^{\text{old}}.$$

It is encrypted to the payee's address  $(d, \text{pk}_d)$ :

$$\begin{aligned} \text{epk} &:= [\text{esk}]_{\text{g}_d}, & K &:= \text{KDF}([\text{esk}] \text{pk}_d, \text{epk}), & C_{\text{enc}} &:= \text{Enc}_K(d, v, \text{rseed}, \text{memo}) \\ \text{ock} &:= \text{PRF}_{\text{ovk}}(\text{cv}^{\text{net}}, \text{cmx}, \text{epk}), & C_{\text{out}} &:= \text{Enc}_{\text{ock}}(\text{pk}_d, \text{esk}) \end{aligned}$$

The payee opens  $C_{\text{enc}}$  with  $\text{ivk}$ ; the payer opens  $C_{\text{out}}$  with  $\text{ovk}$ . Change is an output to an internal address of the payer.

## A payment of 5 ZEC

A note of 7 ZEC pays 5 ZEC; every Action has one spend and one output:

| Action                         | $v^{\text{old}}$ | $v^{\text{new}}$ | $v^{\text{old}} - v^{\text{new}}$ |
|--------------------------------|------------------|------------------|-----------------------------------|
| note $\rightarrow$ payee       | 7 ZEC            | 5 ZEC            | 2 ZEC                             |
| dummy $\rightarrow$ change     | 0                | 1.9999 ZEC       | -1.9999 ZEC                       |
| $v_{\text{balance}}$ (the fee) |                  |                  | 0.0001 ZEC                        |

The value balance  $v_{\text{balance}}$  leaves the shielded pool: here, as the fee.

## What the proof asserts

For public  $(rt, cv^{\text{net}}, nf^{\text{old}}, rk, cmx, \text{flags})$ , the prover knows a witness such that:

1. the old note opens  $cm^{\text{old}}$ , and  $cm^{\text{old}}$  has a valid path to  $rt$  (if  $v^{\text{old}} \neq 0$ );
2.  $nf^{\text{old}} = \text{DeriveNullifier}_{nk}(\rho, \psi, cm^{\text{old}})$ ;
3.  $rk = ak^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}}$ , and  $pk_d^{\text{old}} = [\text{Commit}_{r_{ivk}}^{\text{ivk}}(ak, nk)] g_d^{\text{old}}$ ;
4.  $cv^{\text{net}} = cv(v^{\text{old}} - v^{\text{new}}, rcv)$ ;
5. the new note opens  $cmx$ , with  $\rho^{\text{new}} = nf^{\text{old}}$ ;
6. the flags hold (spends, outputs, cross-address).

One Halo 2 proof for all Actions of the bundle.

# Signatures

Every signature signs SigHash, a digest of the transaction without its proofs, signatures and anchor.

- **Spend authorisation**, per Action: under  $rsk$ , checked against  $rk$ . Only a holder of ask can produce it.
- **Binding**, per bundle:

$$bsk := \sum_i rcv_i, \quad bvk := \sum_i cv_i - [v_{\text{balance}}] V^{\text{Orchard}}$$

under  $bsk$ , checked against  $bvk$ .

# Verification

## What a node checks

| check                           | guarantees                     |
|---------------------------------|--------------------------------|
| the proof, on the public fields | statement 1–6 for every Action |
| $rt$ is a past anchor           | the spent note exists          |
| $nf \notin$ nullifier set       | not spent before               |
| spend signature under $rk$      | the owner authorised it        |
| binding signature under $bvk$   | value is conserved             |

Then: every  $nf$  joins the set, every  $cmx$  joins the tree. No step reads a note.

# Why the binding signature enforces balance

Commitments add:

$$\sum_i cv_i = \left[ \sum_i (v_i^{\text{old}} - v_i^{\text{new}}) \right] V^{\text{Orchard}} + \left[ \sum_i rcv_i \right] R^{\text{Orchard}}$$

so

$$bvk - [bsk] R^{\text{Orchard}} = \left[ \sum_i (v_i^{\text{old}} - v_i^{\text{new}}) - v_{\text{balance}} \right] V^{\text{Orchard}}.$$

A valid signature proves knowledge of  $b$  with  $bvk = [b] R^{\text{Orchard}}$ . If the values did not balance,  $b$  would yield  $\log_{R^{\text{Orchard}}} V^{\text{Orchard}}$ , a relation between two hashed generators: infeasible.

With 64-bit values and fewer than  $2^{16}$  Actions, the sum lies in  $(-2^{80}, 2^{80})$  and  $2^{80} \ll p_{\text{Vesta}}$ : balance modulo  $p_{\text{Vesta}}$  is balance over the integers.

**How Halo 2 proves**

## The claim becomes a table

"I know  $x$  with  $x^3 + x + 5 = 35$ " over  $\mathbb{F}_{97}$ ; the witness  $x = 3$ :

$$3 \xrightarrow{\text{square}} 9 \xrightarrow{\times 3} 27 \xrightarrow{+3} 30 \xrightarrow{+5} 35.$$

One operation per row, one gate equation on every row:

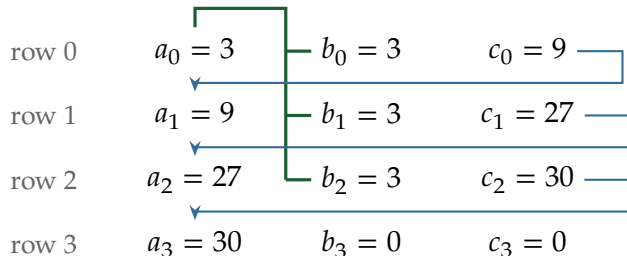
$$q_M a b + q_L a + q_R b + q_O c + q_C + \text{PI} = 0$$

every cell in  $\mathbb{F}_{97}$ .

| row | advice |     |     | fixed |       |       |       |       | instance |                     |
|-----|--------|-----|-----|-------|-------|-------|-------|-------|----------|---------------------|
|     | $a$    | $b$ | $c$ | $q_M$ | $q_L$ | $q_R$ | $q_O$ | $q_C$ | PI       |                     |
| 0   | 3      | 3   | 9   | 1     | 0     | 0     | -1    | 0     | 0        | $x \cdot x = x^2$   |
| 1   | 9      | 3   | 27  | 1     | 0     | 0     | -1    | 0     | 0        | $x^2 \cdot x = x^3$ |
| 2   | 27     | 3   | 30  | 0     | 1     | 1     | -1    | 0     | 0        | $x^3 + x = s$       |
| 3   | 30     | 0   | 0   | 0     | 1     | 0     | 0     | 5     | -35      | $s + 5 = 35$        |

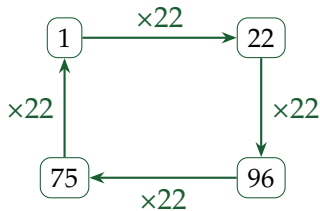
Advice: the prover's secret trace. Fixed: the circuit. Instance: the public 35. Copy constraints wire the rows; range checks are lookups into a table.

## Equal values must be wired together



Equal numbers in a table are not a constraint. These *copy constraints* are enforced separately, by a permutation argument.

## Give the rows multiplicative labels



$$\omega = 22, \quad \omega^2 = 96 = -1, \quad \omega^4 = 1$$

$$H = \{1, 22, 96, 75\} \subset \mathbb{F}_{97}$$

Row  $i$  gets the label  $\omega^i$ : a cycle of field values.

## Rows become polynomials

Row  $i$  lives at  $\omega^i \in H$ . Interpolate each column:

$$a(1) = 3, a(22) = 9, a(96) = 27, a(75) = 30 \implies a(X) = 90 + 61X + 22X^2 + 24X^3$$

$$G(X) := q_M ab + q_L a + q_R b + q_O c + q_C + \text{PI}$$

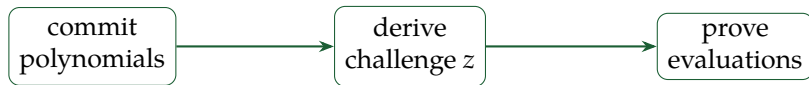
$a, b, c, q_\bullet, \text{PI} \in \mathbb{F}_{97}[X]$  of degree  $< 4$ ;  $G \in \mathbb{F}_{97}[X]$  of degree  $\leq 9$ .

$$Z_H(X) := \prod_{h \in H} (X - h) = X^4 - 1$$

$$\text{every gate holds} \iff G \text{ vanishes on } H \iff Z_H \mid G \iff G = t \cdot Z_H$$

A correct computation *is* a polynomial identity. A wrong one cannot be: the lie  $x^2 = 10$  in row 0 leaves a remainder  $24(1 + X + X^2 + X^3) \neq 0$ .

# One random point



1. The prover *commits* to  $a, b, c$  and  $t$ .
2. The challenge  $z := H(\text{transcript})$  — Fiat–Shamir.
3. The prover opens  $a(z), b(z), c(z), t(z)$ .
4. The verifier checks one equation:

$$G(z) \stackrel{?}{=} t(z) Z_H(z)$$

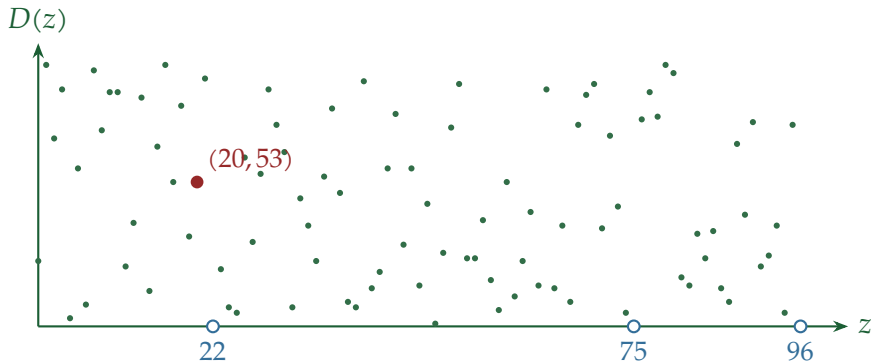
$z \in \mathbb{F}_{97}$  in the toy;  $z \in \mathbb{F}_p$  in the protocol.

Honest,  $z = 20$ :  $G(20) = 75 = 67 \cdot 46$ . The row-0 lie:  $G^*(20) = 20 \neq 64$ .

A false identity is frozen before  $z$  is drawn; it survives only if  $z$  is one of its few roots.

## A nonzero polynomial has few roots

The remainder  $D(X) = 24(1 + X + X^2 + X^3)$  over  $\mathbb{F}_{97}$ : zeros at 22, 75, 96 only.



For uniform  $z$  this cheat passes with probability  $3/97$ ; over the protocol's field, about  $2^{-240}$ .

# Committing to a polynomial

$$P := \langle \vec{a}, \vec{G} \rangle + [r] H, \quad \vec{a} = (a_0, \dots, a_{d-1}), \quad \vec{G} \text{ independent points}$$

$$a(z) = \langle \vec{a}, \vec{b} \rangle, \quad \vec{b} := (1, z, z^2, \dots, z^{d-1})$$

$a_i, r, z \in \mathbb{F}_p$ ;  $\vec{G}, H, P \in \mathcal{U}$ , the Vesta curve, whose scalar field is  $\mathbb{F}_p$ .

- A Pedersen commitment to the coefficient vector: hiding, binding.
- Opening at  $z$  is an inner product with a public vector.
- The problem: convince the verifier of  $\langle \vec{a}, \vec{b} \rangle = v$  in fewer than  $d$  group elements, without sending  $\vec{a}$ .

# The inner-product argument: one fold

Split every vector in half and publish two cross terms:

$$\begin{aligned} L &:= \langle \vec{a}_{\text{lo}}, \vec{G}_{\text{hi}} \rangle + [\ell] H + [\langle \vec{a}_{\text{lo}}, \vec{b}_{\text{hi}} \rangle] U, \\ R &:= \langle \vec{a}_{\text{hi}}, \vec{G}_{\text{lo}} \rangle + [r_R] H + [\langle \vec{a}_{\text{hi}}, \vec{b}_{\text{lo}} \rangle] U. \end{aligned}$$

Challenge  $u$ ; both sides fold:

$$\vec{a}' := u \vec{a}_{\text{lo}} + u^{-1} \vec{a}_{\text{hi}}, \quad \vec{G}' := u^{-1} \vec{G}_{\text{lo}} + u \vec{G}_{\text{hi}}, \quad \vec{b}' := u^{-1} \vec{b}_{\text{lo}} + u \vec{b}_{\text{hi}}$$

$$P + [v] U + [u^2] L + [u^{-2}] R = \langle \vec{a}', \vec{G}' \rangle + [r + u^2 \ell + u^{-2} r_R] H + [\langle \vec{a}', \vec{b}' \rangle] U$$

$$u \in \mathbb{F}_p^\times; L, R, U, H \in \mathcal{U}; \ell, r_R \in \mathbb{F}_p.$$

The same claim at half the length. The verifier folds  $\vec{G}$  and  $\vec{b}$  itself and never sees  $\vec{a}$ .

## The recursion

After  $k = \log_2 d$  rounds every vector has length 1. The proof:  $2k$  points  $(L_j, R_j)$ , one scalar  $a$ , one blinder  $r^*$ .

$$P + [v] U + \sum_{j=1}^k ([u_j^2] L_j + [u_j^{-2}] R_j) \stackrel{?}{=} [a] G^{(0)} + [r^*] H + [a \cdot b^{(0)}] U$$

$a, b^{(0)}, r^*, u_j \in \mathbb{F}_p; G^{(0)}, L_j, R_j \in \mathcal{U}$ .

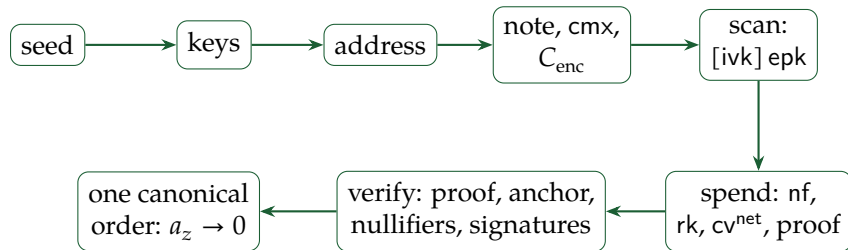
- The scalar  $b^{(0)} = \prod_j (u_j^{-1} + u_j z^{2^{j-1}})$ , with  $u_1$  the last challenge: logarithmic work.
- The point  $G^{(0)} = \langle \vec{s}, \vec{G} \rangle$ : one multi-scalar multiplication, shared across a batch of proofs.
- Binding: two openings would give a discrete-log relation among  $\vec{G}, H$ ; an extractor rewinds three challenges per round.

# The Orchard circuit

- Arithmetic over  $\mathbb{F}_p$ , so Pallas points  $cm, rk, cv$  are native; the polynomial commitments live on Vesta.
- Gadgets: Sinsemilla (commitments, tree), Poseidon (nullifier), elliptic-curve arithmetic, lookups for range checks.
- The instance:  $anchor, cv, nf, rk, cmx, flags$ .
- Generators from a hash: no trusted setup, no ceremony.
- Proof size grows linearly in the number of Actions.

Every clause of the statement is a set of rows; one proof, one random point, one inner-product argument.

# The whole path



The verifier learns that every check passes, the fee, the anchor and the flags, and nothing about the notes.